

Section 3.6 Logarithmic Differentiation

Law of Log's

$$\log(A \cdot B) = \log A + \log B$$

$$\log\left(\frac{A}{B}\right) = \log A - \log B$$

$$\log(A^n) = n \cdot \log A$$

Use the 3 rules to help find derivatives.

Ex ① $f(x) = \ln\left(\frac{\sqrt{x+1}}{x+1}\right)$

If we do NOT use the 3 rules above,
the derivative would look like this.

$$f'(x) = \frac{1}{\frac{\sqrt{x+1}}{x+1}} \cdot \left(\frac{\frac{1}{2\sqrt{x+1}} \cdot (x+1) - \sqrt{x+1} \cdot 1}{(x+1)^2} \right)$$

Then we'd need to do lots of
Algebra to simplify this!

Instead, do the algebra first
And then take the derivative.

$$f(x) = \ln \left(\frac{\sqrt{x+1}}{x+1} \right)$$

$$f(x) = \ln (x+1)^{1/2} - \ln (x+1)$$

$$f(x) = \frac{1}{2} \cdot \ln (x+1) - \ln (x+1)$$

Now take the derivative.

$$f'(x) = \frac{1}{2(x+1)} - \frac{2 \cdot 1}{2 \cdot (x+1)}$$

$$f'(x) = \frac{-1}{2(x+1)}$$

Ex ② $y = \log_7 x$ find $\frac{dy}{dx}$

$$7^y = x$$

Now take derivative on each side.

$$7^y \cdot \ln 7 \cdot y' = 1$$

$$y' = \frac{1}{7^y \cdot \ln 7}$$

replace 7^y with x

$$y' = \frac{1}{x \cdot \ln 7}$$

Ex ③ Find y' for $y = x^x$

Take $\ln$ of both sides.

$$\ln y = \ln(x^x)$$

$$\ln y = x \cdot \ln x$$

Take derivative of each side.

$$\frac{1}{y} \cdot y' = 1 \cdot \ln x + x \cdot \frac{1}{x}$$

$$y' = y (\ln x + 1)$$

Replace y with x^x

$$y' = x^x (\ln x + 1)$$

Ex ④ $y = \frac{\sqrt{(x+3)^7} \cdot (5x+1)^4}{(2x-6)^{10}}$

Take $\ln$ of both sides.

$$\ln y = \ln \left[\frac{(x+3)^{7/2} \cdot (5x+1)^4}{(2x-6)^{10}} \right]$$

Use the log rules to separate the right side into 3 logarithms.

$$\ln y = \ln (x+3)^{7/2} + \ln (5x+1)^4 - \ln (2x-6)^{10}$$

Now move the exponents in front
and multiply.

$$\ln y = \frac{7}{2} \cdot \ln(x+3) + 4 \cdot \ln(5x+1) - 10 \cdot \ln(2x-6)$$

Now, take the derivatives.

$$\frac{y'}{y} = \frac{7}{2} \cdot \left(\frac{1}{x+3} \right) + 4 \cdot \left(\frac{5}{5x+1} \right) - 10 \cdot \left(\frac{2}{2x-6} \right)$$

Multiply both side by y and simplify the fractions.

$$y' = y \cdot \left[\frac{7}{2(x+3)} + \frac{20}{5x+1} - \frac{10}{x-3} \right]$$

Finally, replace y with the original function.

$$y' = \frac{\sqrt{(x+3)^7} \cdot (5x+1)^4}{(2x-6)^{10}} \cdot \left[\frac{7}{2(x+3)} + \frac{20}{5x+1} - \frac{10}{x-3} \right]$$